

## ■ Database Availability

-- oel7v9 서버에서 진행

= Incremental Backup 수행

```
RMAN> BACKUP INCREMENTAL LEVEL 0 DATABASE;
```

```
SQL> create table hr.emp8 tablespace users as select * from dba_objects;
```

Table created.

```
SQL> insert into hr.emp8 select * from hr.emp8;
```

72596 rows created.

```
SQL> commit;
```

Commit complete.

```
RMAN> backup incremental level 1 database;
```

```
SQL> create table hr.emp9 tablespace users as select * from dba_objects;
```

Table created.

```
SQL> insert into hr.emp9 select * from hr.emp8;
```

145192 rows created.

```
SQL> commit;
```

Commit complete.

```
RMAN> BACKUP INCREMENTAL LEVEL 1 CUMULATIVE DATABASE;
```

```
RMAN> list backup of database;
```

```
RMAN> BACKUP VALIDATE CHECK LOGICAL DATABASE ARCHIVELOG ALL;
```

- Change Tracking

```
SQL> ALTER DATABASE ENABLE BLOCK CHANGE TRACKING USING FILE  
'/home/oracle/backup/change_tracking.ctf';
```

Database altered.

```
SQL> !ps -ef|grep orcl
```

```
oracle    5032      1  0 09:54 ?          00:00:02 ora_pmon_orcl  
oracle    21752     1  0 14:21 ?          00:00:00 ora_ctwr_orcl  
oracle    21769 17559  0 14:21 pts/1    00:00:00 /bin/bash -c ps -ef|grep orcl  
oracle    21771 21769  0 14:21 pts/1    00:00:00 grep orcl
```

```
SQL> SELECT status, filename from v$block_change_tracking;
```

| STATUS | FILENAME |
|--------|----------|
|--------|----------|

|         |                                         |
|---------|-----------------------------------------|
| ENABLED | /home/oracle/backup/change_tracking.ctf |
|---------|-----------------------------------------|

- 문제 1 : brlab7.sh 를 수행후 Autobackup 기능을 사용하여 DB 를 복구하기
  
- 문제 2 : brlab8.sh 를 수행후 Level 0 백업과 Level 1 누적(Cumulative) 백업을 활용하여 최소한의 시간으로 복구하기
  
- 문제 3 : BACKUP VALIDATE 당시의 SCN 으로 복구하기